

U M O L O G Y

Quantum physics is at the heart of many established and emerging technologies. Applications that harness quantum principles in their design—ranging from everyday electronics to satellite timekeepers—can be broadly described as quantum technologies. The devices that laid the foundation of the Information Age are built on quantum physics—for instance, semiconductor devices are designed around the quantization of electron energy levels, which restrict the movement of charge in a lattice. In the near future, similarly transformative new technologies could be built on other quantum phenomena, such as entanglement.